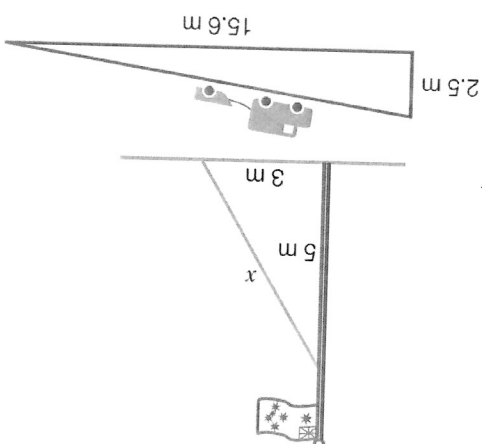
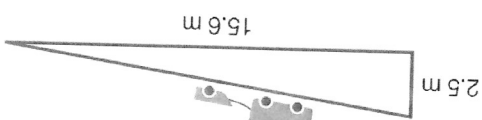


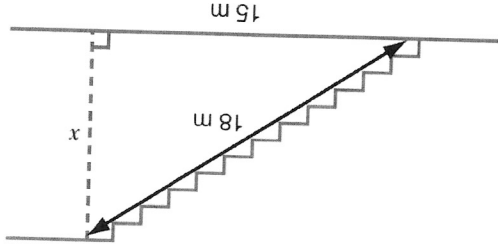
- 1 (a) A support wire is attached 5 m up a flagpole. The other end of the wire is attached to the ground 3 m from the base of the flagpole. What is the length of the wire?



- (b) A boat ramp is to be built that will cover a horizontal distance of 15.6 metres while descending a vertical distance of 2.5 metres. What is the length of the ramp?



- 2 The escalator at a department store is 18 m long. When you ride on the escalator you move across 15 m. Calculate the height through which you travel from the lower level to the higher level.



- 3 (a) A leaf rake 1.8 m long is leaning against a vertical post. The distance from the bottom of the post to the bottom of the rake is 0.6 m. How high up the post will the top of the rake be?

- (b) A rope is tied to a vertical tree trunk 2 m from the ground and is attached to the ground at a point 1.2 m from the base of the tree. How long is the segment of rope between the tree and the ground?

- 4 The top of a 6.5 m long ladder is resting against a wall at a point that is 4.2 m above the ground. What is the approximate distance of the base of the ladder from the base of the wall?

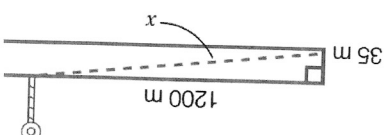
- A 2.46 m B 2.99 m C 3.24 m D 4.96 m

- 5 A slide is 5 m long and the vertical ladder is 3.6 m. How far is the base of the ladder from the bottom of the slide?

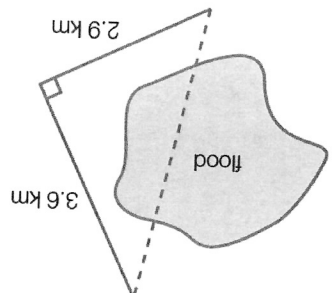
- 6 Andrew is standing at the corner of a rectangular park, 35 m long and 28 m wide. He has decided to cross diagonally rather than go around the outside.

- (a) How far does Andrew travel when crossing the park?
(b) How much distance does he save by not going around the outside?

- 7 Kensington race course has races that are run over a straight 1200 m course. One horse starts near the outside fence and runs in a straight line towards the finishing post on the inside fence. If the track is 35 m wide, how much further does this horse run than a horse that runs in a straight line along the inside fence?



- 8 During a heavy rainstorm, the road that Allie normally takes to travel home from school has become flooded. She must take a detour as shown in the diagram. How much further does she now have to travel?



Hint Draw a diagram for Questions 5 and 6 to represent the given information.